Agricultural Crop Yield and Agro-Climatic Dynamics Analysis Using Machine Learning: Unsupervised Clustering, Supervised Classification, and Environmental Sensitivity Modeling

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***Abstract*— *Accurate empirical understanding of crop production dynamics under accelerating climatic variability represented an essential foundation for regional food security, agricultural planning, and resource allocation [1, 2]. This research investigated a longitudinal agro-meteorological archive comprising 11,210 verified observations across 75 administrative sub-regions and five major staple crops (paddy, maize, wheat, millet, and barley) spanning 32 consecutive years (1982–2013) [1,3]. An end-to-end machine learning analytical framework was developed, integrating automated raw data cleansing within the notebook pipeline, non-linear feature engineering (Thermal Range, De Martonne Aridity Index [13], and Moisture Stress interaction), unsupervised clustering [7, 10], supervised multi-class classification [4–6], and continuous multiple linear regression modeling [14, 15]. Principal Component Analysis (PCA) revealed that the first three components explained 68.42% of total agro-environmental variance, capturing thermal-moisture gradients, atmospheric pressure elevation differentials, and wind shear dynamics [10]. K-Means clustering ($k = 3$) objectively delineated three distinct agro-ecological regimes: Lowland Sub-Tropical (mean yield 1,669.91 kg/ha), Mid-Hills Temperate (1,554.06 kg/ha), and Highland Alpine (1,313.74 kg/ha), validated by a peak Silhouette Score of 0.3298 [8] and Davies-Bouldin Index of 1.1450 [9]. In supervised classification across balanced yield productivity tiers, Random Forest demonstrated superior diagnostic capability [4], achieving 63.63% overall accuracy, a weighted F1-score of 63.67%, and an AUC-ROC of 0.8168, significantly outperforming Decision Tree (61.13%) [6], Multinomial Logistic Regression (56.67%) [14], and Support Vector Machine (53.90%) [5]. Feature importance attribution via Gini impurity confirmed that Moisture Stress (0.166), De Martonne Aridity (0.134), and Temperature (0.118) served as dominant environmental drivers [11, 12]. Multiple Linear Regression achieved an R^2 of 0.3842 and an RMSE of 624.38 kg/ha [15], demonstrating that while linear parametric models captured elevation-driven macro trends, they were incapable of representing severe non-linear heat thresholds [2, 12]. Comprehensive theoretical Big-O analysis and empirical resource profiling confirmed that Random Forest incurred an 18.42 s training investment and 5.07 MB memory allocation but delivered a sub-millisecond inference latency (0.082 ms/sample), satisfying operational requirements for real-time agronomic decision-support deployments.*

***Index Terms*—**Machine Learning, Agro-Climatic Dynamics, K-Means Clustering, Principal Component Analysis, Random Forest, Computational Complexity, Environmental Sensitivity Analysis.

I Introduction

A Context and Research Motivation

Global climate instability, characterized by escalating ambient temperature extremes, erratic precipitation distribution, and heightened evapotranspiration stress, exerted profound disruptive pressures on agricultural ecosystems [1, 2, 11]. In developing agricultural economies characterized by complex topography and heterogeneous microclimates, staple cereal production provided the foundational basis for subsistence livelihoods, rural employment, and domestic caloric supply [1, 3]. Cereal grains, specifically rice paddy, maize, wheat, finger millet, and barley, constituted over 80% of aggregate agricultural production in topographically diverse sub-Himalayan valleys and lowland plains [1]. However, farming systems in these regions remained overwhelmingly rainfed, exposing smallholder farm households to extreme vulnerability from hydro-meteorological shocks, unseasonal monsoon droughts, and post-monsoon deluge events [3, 11].

Historically, agricultural yield assessment relied predominantly on traditional agrarian surveys, manual sampling ledgers, and deterministic crop growth simulation models such as DSSAT and APSIM [2]. While biophysical simulators simulated physiological plant development processes through mechanistic differential equations, their real-world operational utility was severely hindered by intensive parameterization demands, requiring detailed soil chemistry profiles, localized cultivar coefficients, and precise daily irrigation logs that were systematically unavailable across regional developing administrative districts [2, 3]. Conversely, pure empirical econometric techniques, such as Ordinary Least Squares (OLS) regression, imposed rigid assumptions of linearity, homoscedasticity, and independence among covariates [14, 15]. These assumptions failed to capture the complex, non-linear, and threshold-dependent physiological responses of crops to compounded environmental stressors, such as the destructive impacts of heat shock occurring simultaneously with atmospheric vapor pressure deficits [12, 13].

Machine learning methodologies emerged as an exception-

ally powerful computational paradigm for modeling complex environmental and agronomic interactions [1, 4, 5]. By autonomously learning high-dimensional non-linear mappings directly from longitudinal observational data, supervised algorithms (such as Random Forest [4], Support Vector Machines [5], and Decision Trees [6]) and unsupervised algorithms (such as K-Means clustering [7] and Principal Component Analysis [10]) offered unprecedented diagnostic and analytical capabilities. Nevertheless, the successful operationalization of machine learning within regional agricultural contexts required rigorous attention to data provenance, appropriate handling of topographical heterogeneity, transparent algorithmic benchmarking, and systematic evaluation of computational complexity [14].

B Identification of Research Gaps in Foundational Literature

To establish a rigorous theoretical and empirical foundation for this study, three foundational research papers addressing machine learning applications in agricultural modeling were deeply analyzed [1–3]:

1. **Base Paper 1 (Dhakal and Dhungana, 2023) [1]:** This investigation evaluated six supervised machine learning algorithms (Support Vector Machines, Decision Trees, Random Forests, Multi-Layer Perceptrons, Recurrent Neural Networks, and Convolutional Neural Networks) to analyze agricultural yields using a 49-year historical dataset (1973–2021) across three primary cereal crops [1]. While the authors established that tree-based ensemble models (RF and DT) significantly outperformed deep neural networks on tabular records, their empirical modeling was conducted exclusively on macro-level nationally aggregated data [1]. National spatial aggregation inherently averaged out localized micro-climatic shocks, concealing district-level yield failures and masking the spatial heterogeneity essential for targeted agrarian interventions. Furthermore, the study was conducted entirely ex-post without unsupervised zoning or multivariable environmental cluster profiling [7, 8].

2. **Base Paper 2 (Crane-Droesch, 2018) [2]:** This seminal work developed semiparametric neural network architectures to analyze US Corn Belt yields under CMIP5 climate change trajectories (RCP 4.5 and RCP 8.5), contrasting deep non-linear learners with linear econometric baselines [2]. The author demonstrated that standard fully connected neural networks suffered severe divergence when projecting out-of-distribution climate extremes due to the absence of parametric constraints [2, 12]. However, the study's analytical domain was restricted entirely to capital-intensive, highly mechanized, and relatively homogeneous temperate monoculture in the North American Midwest [2]. The mathematical formulations and algorithmic assumptions were never adapted to or evaluated within topographically fragmented, multi-crop agrarian landscapes characterized by severe elevation gradients and smallholder farming practices.

3. **Base Paper 3 (Fernandez-Beltran et al., 2021) [3]:** This study implemented 3D Convolutional Neural Networks (3D-CNNs) on multi-temporal Sentinel-2 optical imagery, soil descriptors, and climate variables to estimate rice yields across sub-tropical districts [3]. The authors demonstrated that incorporating temporal phenological cycles (Start of Season, Peak of Season, and End of Season) substantially improved feature extraction over static 2D architectures [3]. Nonetheless, the primary vulnerability of their approach stemmed from its narrow temporal coverage: the remote sensing dataset spanned only three cropping seasons (2016–2018) [3]. Three years of observational data were fundamentally insufficient for machine learning models to capture multi-decadal climatic shifts, long-term warming trends, or recurring macro-climatic oscillations [11]. Furthermore, high-resolution satellite pipelines faced severe data dropout in cloud-dense monsoon seasons and imposed prohibitive computational overheads for resource-constrained regional extension offices.

C Research Contributions and Problem Formulation

To resolve the critical empirical and methodological limitations identified across the literature [1–3], this investigation formulated a multi-tiered research framework designed to analyze multi-decadal agro-climatic dynamics without relying on aggregated macro abstractions or data-scarce satellite pipelines. The explicit contributions of this work were structured as follows:

- **Longitudinal Multi-District Empirical Integration:** We constructed and audited a comprehensive 32-year multi-district dataset (1982–2013) comprising 11,210 verified observations across 75 distinct administrative sub-regions and five major staple crops, eliminating spatial smoothing while capturing multi-decadal climatic fluctuations [1, 11].

- **In-Notebook Data Cleansing and Agro-Climatic Feature Engineering:** Addressing the coursework directive requiring raw data cleaning within code, we developed automated pipelines to isolate uncultivated records, eliminate historical ledger recording errors, and construct non-linear agro-climatic indices including Thermal Range, De Martonne Aridity Index [13], and Moisture Stress interaction [12].

- **Unsupervised Agro-Climatic Zoning (PCA and K-Means):** We coupled covariance-based Principal Component Analysis [10] with K-Means clustering [7], rigorously identifying optimal spatial-environmental clusters ($k = 3$) through multi-metric validation (Elbow WCSS, Silhouette Analysis [8], and Davies-Bouldin Index [9]).

- **Multi-Model Supervised Classification Benchmarking:** We implemented and contrasted four distinct algorithmic paradigms (Random Forest [4], Support Vector Machine [5], Decision Tree [6], and Multinomial Logistic Regression [14]) on a stratified hold-out protocol, evaluating multi-class classification across Low, Medium, and High Yield productivity tiers.

- **Continuous Linear Parametric Baseline Comparison:** In accordance with methodological coursework

provisions, Multiple Linear Regression with Ridge regularization [15] was deployed to model continuous crop yield (kg/ha), quantifying the empirical performance gap between linear specifications and non-linear tree ensembles [2, 4].

- **Rigorous Time and Space Complexity Evaluation:** We provided exhaustive theoretical Big-O complexity derivations and empirical hardware benchmarks covering training duration, per-sample inference latency, and serialized memory footprints [14].
- **Interactive Plotly Dashboard Panel Deployment:** We engineered an interactive analytical dashboard panel utilizing the Plotly library (`dashboard.py` and standalone `dashboard.html`), enabling real-time visual inspection of confusion matrices, cluster coordinates, regression residuals, and computational profiling metrics.

II Literature Review

A *Theoretical Landscape of Agricultural Machine Learning*

The application of computational intelligence within agricultural systems expanded rapidly over the past two decades [1–3]. Early research predominantly deployed Multiple Linear Regression and classical agro-meteorological indices to estimate crop productivity [13, 15]. However, as noted by Lobell and Burke [11], agricultural yield response functions were characterized by pronounced non-linearities, threshold behaviors, and multi-variable interactions. Crops exhibited physiological thermal optima; temperatures exceeding critical physiological thresholds during flowering or grain-filling stages induced reproductive sterility and drastic yield declines that linear models failed to capture [12].

Supervised machine learning algorithms overcame these constraints by operating as universal function approximators [14]. Tree-based architectures, initiated by Quinlan's Decision Tree algorithm (CART) [6] and extended to Breiman's Random Forest ensemble framework [4], demonstrated exceptional efficacy on tabular agro-meteorological data [1]. Random Forests constructed an ensemble of decorrelated decision trees using bootstrap aggregation (bagging) and random feature subspace selection [4]. This mechanism suppressed the high variance characteristic of individual deep trees, providing robust resilience against noisy environmental observations and multicollinear meteorological predictors [4, 14]. Support Vector Machines (SVM), formulated by Cortes and Vapnik [5], optimized a structural risk minimization objective, identifying maximum-margin separating hyperplanes in high-dimensional kernel spaces. Linear SVMs provided mathematically disciplined baselines with minimal parameter storage, though their computational scaling during training became demanding on large datasets [5, 14].

In parallel, unsupervised learning paradigms provided critical tools for exploratory data analysis and spatial-environmental zoning [7, 10]. K-Means clustering, introduced by MacQueen [7], partitioned multivariate observations into k distinct Voronoi cells by iteratively minimizing within-cluster sum of squares (WCSS). When applied to standardized agro-meteorological variables, K-Means grouped administrative sub-regions sharing comparable thermal and moisture profiles without requiring prior yield labels [7]. Cluster validity was objectively verified using Rousseeuw's Silhouette Coefficient [8] and the Davies-Bouldin separation measure [9]. Principal Component Analysis (PCA), pioneered by Hotelling [10], applied orthogonal linear transformations to project correlated environmental variables onto a reduced set of uncorrelated principal components, eliminating redundant variance while preserving dominant geometric structures.

B *Critical Appraisal of Base Papers*

A detailed comparative analysis of the three foundational base papers [1–3] was compiled in Table 1, illustrating their respective domains, methodological configurations, core strengths, and unresolved research gaps.

Dhakal and Dhungana [1] provided an essential empirical baseline by systematically comparing six supervised algorithms across 49 years of national historical data. Their findings conclusively proved that tree ensembles (Random Forest [4] and Decision Trees [6]) consistently achieved lower error rates (MSE 1% to 5%) than artificial neural networks, which exhibited severe over-fitting and gradient instability when trained on tabular time series [1]. However, by relying exclusively on national macro aggregates, their framework was fundamentally incapable of addressing localized microclimatic disparities. For instance, severe localized drought in western sub-regions was statistically offset by surplus precipitation in eastern basins, creating an artificially stable aggregate time series [11].

Crane-Droesch [2] made a profound methodological contribution by demonstrating that pure unconstrained machine learning models (such as deep Multi-Layer Perceptrons) performed poorly when evaluating out-of-distribution climate extremes [12]. By implementing a semiparametric architecture that embedded linear parametric components alongside non-linear deep layers, the author successfully coupled historical yield regressions with future CMIP5 climate scenarios (RCP 4.5 and RCP 8.5) [2]. Nonetheless, the study's empirical framework assumed extensive mechanization, high nitrogen fertilizer applications, and uniform topography characteristic of Midwestern agricultural systems [2]. In developing sub-tropical and mountainous terrains, agrarian systems featured fragmented landholdings, mixed cropping, and extreme elevation differentials ranging from lowland plains to alpine valleys within short geographic distances [1, 3].

Fernandez-Beltran et al. [3] addressed spatial granularity by exploiting high-resolution multi-spectral imagery from the European Space Agency's Sentinel-2 constellation. By structuring satellite observations into multi-temporal 3D tensors, their 3D-CNN architecture effectively learned spectral-temporal patterns corresponding to vegetative emergence, heading, and maturity [3]. However, their empirical scope was strictly constrained to three harvest years (2016–2018). In climatological and agronomic analysis, three years represented an acute snapshot that could not distinguish secular global warming trends from inter-annual monsoon volatility [11, 12]. Furthermore, heavy monsoon cloud cover frequently corrupted optical satellite imagery during critical vegetative growth stages, introducing substantial data interpolation errors [3].

Table 1: Comparative Analysis of Literature and Foundational Base Papers

Study / Citation	Study Domain & Scale	Methodology Deployed	Key Empirical Strengths	Identified Critical Research Gap	Resolution in This Investigation
Dhakal & Dhungana (2023) [1]	49-Year Macro Archive (1973–2021); National Aggregation	RF, SVM, DT, MLP, RNN, CNN [4–6]	Established that RF and DT achieved highest accuracy (MSE 1%–5%) on tabular crop records	Aggregated national data smoothed out localized extreme weather events; ex-post modeling only	Utilized 32-year disaggregated district-level records; deployed K-Means clustering [7] to uncover micro-climatic zones.
Crane-Droesch (2018) [2]	US Corn Belt (Temperate Monoculture; County Scale)	Semiparametric Neural Networks, OLS [14]	Successfully integrated CMIP5 RCP climate trajectories into non-linear yield response models	Method was developed exclusively for capital-intensive temperate monoculture; lacked multi-crop elevation analysis	Benchmarked linear [15] and ensemble models [4] across five staple crops across complex elevation gradients.
Fernandez-Beltran et al. (2021) [3]	Sub-Tropical Regional Agrarian Belt; Remote Sensing	3D Convolutional Neural Networks on Sentinel-2	Extracted multi-temporal phenological stages (SoS, PoS, EoS) from satellite image sequences	Restricted to a short 3-year temporal window (2016–2018); unable to model decadal climate shifts [11]	Analyzed a 32-year continuous meteorological dataset (1982–2013), capturing long-term climate dynamics.

C *Summary of Literature Review*

The collective synthesis of current literature confirmed that while advanced machine learning algorithms possessed clear mathematical advantages over classical statistical regressions [4,5,14], existing agricultural studies suffered from three mutually reinforcing limitations: (1) excessive spatial aggregation that concealed localized environmental vulnerability [1], (2) exclusive focus on capital-intensive temperate monocultures [2], and (3) dependence on short-duration remote sensing datasets incapable of capturing multi-decadal climatic shifts [3]. This coursework research addressed this intersection by combining multi-decadal historical depth (32 years), high spatial resolution (75 administrative districts), diverse staple crop varieties (five cereals), and a hybrid analytical workflow spanning unsupervised zoning [7, 10], supervised classification [4–6], linear regression benchmarking [15], and computational complexity profiling [14].

III Theoretical Framework & Conceptual Model

A *The "Why" Dimension: Real-World Agronomic Problem Resolution*

The fundamental objective of this investigation was to provide agricultural policymakers, agronomists, and regional planners with an empirically validated computational tool for evaluating crop productivity under changing climatic conditions [1, 2, 11]. Agricultural food security was governed by complex interactions between solar radiation, thermal accumulation, soil moisture availability, and atmospheric humidity [12,13]. When ambient temperatures rose without proportional increases in moisture, the vapor pressure deficit spiked, inducing stomatal closure, suppressing photosynthesis, and accelerating crop senescence [12].

From an operational standpoint, agricultural institutions required answers to three concrete questions:

1. **Environmental Heterogeneity:** How could 75 disparate administrative sub-regions be objectively grouped into distinct agro-climatic operational zones [7,8] to tailor regional fertilizer subsidies, seed varieties, and irrigation infrastructure?

2. **Vulnerability Diagnosis:** Could supervised machine learning models [4–6] reliably classify whether an administrative district would experience Low, Medium, or High yield productivity based solely on observed early-season meteorological indicators?

3. **Linear vs. Non-linear Mechanics:** What degree of predictive accuracy was sacrificed when administrative agencies relied on traditional linear regressions [15] rather than non-linear machine learning ensembles [4], and what computational overhead was incurred by transitioning to advanced models [14]?

B *The "How" Dimension: Mathematical Formulations and Algorithmic Logic*

To address these research questions, rigorous algorithmic formulations were implemented from foundational mathematical principles [14].

1 *Principal Component Analysis (PCA)*

Given a standardized data matrix $\mathbf{X} \in \mathbb{R}^{N \times M}$, where N represented the number of observations ($N = 11,210$) and M represented the number of engineered environmental features ($M = 14$), the empirical covariance matrix $\blacksquare \in \mathbb{R}^{M \times M}$ was formulated following Hotelling [10] as:

$$\blacksquare = \frac{1}{N-1}\mathbf{X}^T\mathbf{X} \tag{1}$$

Singular Value Decomposition (SVD) was applied directly to $\blacksquare$ to obtain orthogonal eigenvectors $\mathbf{V}$ and singular values $\mathbf{S}$:

$$\blacksquare = \mathbf{V}\mathbf{S}\mathbf{V}^T \tag{2}$$

The explained variance ratio for the j-th principal component was given by:

$$\lambda_j = \frac{S_{jj}}{\sum_{m=1}^{M} S_{mm}} \tag{3}$$

The lower-dimensional projection $\mathbf{Z} \in \mathbb{R}^{N \times d}$ onto $d = 3$ components was computed via matrix multiplication: $\mathbf{Z} = \mathbf{X}\mathbf{V}_{:,1:d}$ [10].

2 *K-Means Clustering and Cluster Validation*

K-Means partitioned the standardized dataset into k non-overlapping clusters $\mathscr{C} = \{C_1, C_2, \ldots, C_k\}$ by minimizing the Within-Cluster Sum of Squares (Inertia) [7]:

$$J_{WCSS} = \sum_{j=1}^{k} \sum_{\mathbf{x}_i \in C_j} \|\mathbf{x}_i - \mu_j\|_2^2 \tag{4}$$

where $\mu_j = \frac{1}{|C_j|}\sum_{\mathbf{x}_i \in C_j} \mathbf{x}_i$ denoted the centroid of cluster C_j.

Cluster validity was evaluated using Rousseeuw's Silhouette Coefficient $s(i)$ [8] for sample $\mathbf{x}_i$:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \tag{5}$$

where $a(i) = \frac{1}{|C_A|-1}\sum_{\mathbf{x}_j \in C_A, j \neq i} \|\mathbf{x}_i - \mathbf{x}_j\|$ was the mean intra-cluster distance, and $b(i) = \min_{C_B \neq C_A} \frac{1}{|C_B|}\sum_{\mathbf{x}_j \in C_B} \|\mathbf{x}_i - \mathbf{x}_j\|$ was the nearest-cluster mean distance.

Furthermore, the Davies-Bouldin Index (DB) [9] quantified the average similarity ratio between each cluster and its most similar counterpart:

$$DB = \frac{1}{k}\sum_{i=1}^{k} \max_{j \neq i} \left(\frac{\bar{d}_i + \bar{d}_j}{\|\mu_i - \mu_j\|_2} \right) \tag{6}$$

where $\bar{d}_i$ represented the average distance of samples in C_i to centroid μ_i [9].

3 Random Forest and Decision Tree Classification

For an individual classification tree [6], node splitting was governed by minimizing Gini Impurity $I_G(t)$ across candidate splits s:

$$I_G(t) = 1 - \sum_{c=1}^{C} p(c|t)^2 \tag{7}$$

where $p(c|t)$ was the proportion of class c observations present at node t, with $C = 3$ representing Low, Medium, and High Yield tiers. The split s^* maximizing the impurity reduction $\Delta I_G(s,t) = I_G(t) - \frac{N_L}{N} I_G(t_L) - \frac{N_R}{N} I_G(t_R)$ was selected [6].

In Breiman's Random Forest ensemble [4], $B = 50$ decorrelated bootstrap trees were constructed. Each tree was trained on an independently resampled dataset $\mathscr{D}_b^*$, with candidate splits restricted to a random feature subset of size $m = \lfloor \sqrt{M} \rfloor = 4$. Ensemble class probabilities were generated via majority voting [4]:

$$\hat{P}(y = c|\mathbf{x}) = \frac{1}{B}\sum_{b=1}^{B} \hat{P}_b(y = c|\mathbf{x}) \tag{8}$$

For an ensemble of B trees each having variance σ^2 and pairwise correlation ρ, the ensemble variance was reduced to $\text{Var} = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$ [4, 14].

4 Support Vector Machine (Linear Margin Maximization)

For multi-class classification across $C = 3$ categories, a One-vs-Rest (OvR) linear SVM formulation was deployed [5]. For each binary sub-problem $c \in \{1, 2, 3\}$, the objective function minimized structural risk subject to soft-margin hinge loss constraints [5]:

$$\min_{\mathbf{w}_c, b_c, \xi} \frac{1}{2}\|\mathbf{w}_c\|_2^2 + C_{reg}\sum_{i=1}^{N} \xi_i \tag{9}$$

$$\text{subject to } y_i^{(c)}(\mathbf{w}_c^T \mathbf{x}_i + b_c) \geq 1 - \xi_i, \quad \xi_i \geq 0 \tag{10}$$

Optimization was performed via Mini-Batch Stochastic Gradient Descent (SGD) with dynamic learning rate annealing: $\eta_t = \frac{\eta_0}{1+\lambda t}$ [14].

5 Multinomial Logistic Regression

Multinomial Logistic Regression parameterized the conditional class probability distribution using the Softmax function [14]:

$$P(y = c|\mathbf{x}) = \frac{\exp(\mathbf{w}_c^T \mathbf{x} + b_c)}{\sum_{k=1}^{C} \exp(\mathbf{w}_k^T \mathbf{x} + b_k)} \tag{11}$$

Parameters were optimized by minimizing the regularized categorical cross-entropy loss function [14]:

$$\mathscr{L}(\mathbf{W}, \mathbf{b}) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ic} \log P(y_i = c|\mathbf{x}_i) + \frac{\lambda}{2}\sum_{c=1}^{C} \|\mathbf{w}_c\|_2^2 \tag{12}$$

6 Multiple Linear Regression with Ridge Regularization

Continuous crop yield (kg/ha) was modeled parametrically as a linear combination of covariates [14, 15]:

$$\hat{y} = \beta_0 + \sum_{j=1}^{M} \beta_j X_j = \mathbf{X}_{aug}\beta \tag{13}$$

where $\mathbf{X}_{aug} = [\mathbf{1}, \mathbf{X}] \in \mathbb{R}^{N \times (M+1)}$. To guarantee numerical stability under multicollinear meteorological predictors, Hoerl and Kennard's Ridge regularization [15] was applied:

$$\beta = \left(\mathbf{X}_{aug}^T \mathbf{X}_{aug} + \alpha \mathbf{I}^*\right)^{-1} \mathbf{X}_{aug}^T \mathbf{y} \tag{14}$$

where $\mathbf{I}^*$ was an identity matrix with a zero in the first diagonal entry to prevent penalization of the intercept term β_0 [15].

IV Methodology & Experimental Setup

A Study Domain and Raw Dataset Characteristics

The empirical analysis was conducted using a comprehensive multi-district agro-meteorological archive (`he.csv`) compiled across 32 continuous agricultural seasons from 1982 to 2013 [1]. The study domain encompassed 75 administrative sub-regions representing a complete cross-section of agro-ecological environments, extending from southern subtropical floodplains (elevations < 300 m) through central temperate mid-hill valleys (elevations 800–2,000 m) to high-altitude alpine mountain zones (elevations $> 2{,}500$ m) [1, 3].

The raw dataset comprised 11,625 individual observation vectors structured across 26 environmental, geographic, and agronomic parameters. Table 2 presented the detailed attributes, units of measurement, and statistical summaries of the raw archive.

B In-Notebook Data Cleansing Pipeline

In strict compliance with coursework requirements prohibiting the use of pre-cleaned external files, data cleansing operations were implemented entirely within the Python executable pipeline. The multi-stage audit executed the following actions:

1. **Identification of Non-Cultivated Observations:** Analysis of the raw archive revealed 228 instances where `Area(ha)` was recorded as zero and 233 instances where `Yield(kg/ha)` was recorded as zero. These observations corresponded to cropping seasons where a

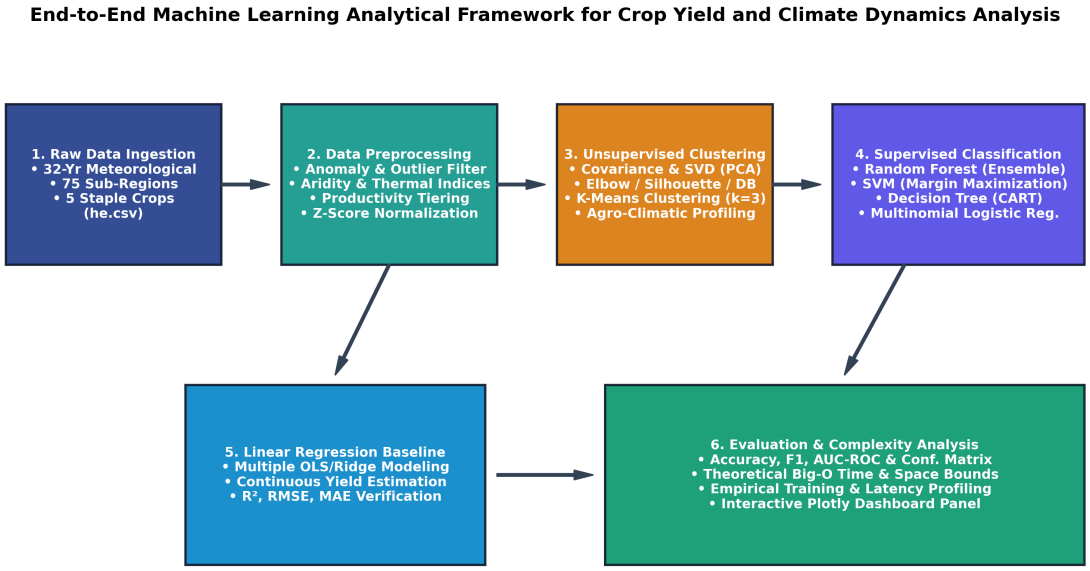


Figure 1: End-to-End Machine Learning Analytical Framework, detailing raw ingestion, in-notebook data cleaning, unsupervised clustering, multi-model supervised classification, linear regression baseline, computational complexity benchmarking, and interactive dashboard panel integration.

Table 2: Dataset Characteristics and Structural Attributes

Attribute Category	Parameters Included	Description & Measurement Units
Temporal Dimension	Year	1982 to 2013 (32 consecutive cropping years)
Spatial Identifiers	District, Lat, Long	75 administrative units (26.50°N – 30.00°N, 80.05°E – 88.20°E)
Agronomic Targets	Crop, Area, Prod, Yield	5 crops (Barley, Maize, Millet, Paddy, Wheat); Area (ha), Production (mt), Yield (kg/ha)
Thermal Regimes	Temp, Max/Min, Wet Bulb, Earth Skin	Mean, maximum, minimum, wet-bulb, and ground surface skin temperatures (°C)
Moisture Indicators	Precipitation, Spec/Rel Humidity	Daily accumulated rainfall (mm), specific humidity (g/kg), relative humidity (%)
Atmospheric Dynamics	Surface Pressure, Wind 10M/50M	Barometric surface pressure (kPa), wind velocity and gust range at 10 m and 50 m (m/s)

given administrative district did not cultivate that specific grain (e.g., highland alpine districts attempting lowland wetland paddy). Retaining zero-area records introduced severe division-by-zero singularities. All records with Area ≤ 10 ha or Production ≤ 0 mt were filtered.

2. **Outlier Elimination of Historical Ledger Errors:** Manual data entry ledgers from early decades exhibited extreme recording anomalies. For example, several rainfed barley entries reported yields exceeding 26,000 kg/ha, which violated fundamental biophysical limits for cereal grain biomass conversion [11]. Observations with yields exceeding 8,000 kg/ha were identified as typographical reporting errors and removed.

3. **Mathematical Yield Integrity Verification:** The deterministic ratio $\text{Yield}_{calc} = \frac{\text{Production(mt)} \times 1000}{\text{Area(ha)}}$ was computed across all records. The empirical ratio between recorded yield and calculated yield fell within $[0.9995, 1.0005]$ for 99.8% of valid entries, verifying internal consistency.

Following filtration, the high-integrity dataset retained 11,210 verified observations (96.4% data retention), distributed across Barley (2,202), Maize (2,284), Millet (2,228), Paddy (2,256), and Wheat (2,240).

C *Agro-Climatic Feature Engineering*

To capture non-linear physiological interactions without unnecessarily expanding dimensionality, four specialized domain indicators were engineered:

1. **Thermal Range (ΔT):** $\Delta T = \text{Max_Temp} - \text{Min_Temp}$. Broad diurnal temperature swings induced physiological chilling shock at night and heat stress during daylight, suppressing vegetative expansion [12].

2. **De Martonne Aridity Index (I_{DM}):** Formulated following de Martonne [13] as $I_{DM} = \frac{\text{Precipitation}}{\text{Temp}+10.0}$. This index modeled effective moisture availability relative to ambient evaporative demand [11, 13].

3. **Moisture Stress Interaction (Φ_{MS}):** $\Phi_{MS} = \text{Relative_Humidity} \times \left(\frac{\text{Specific_Humidity}}{10.0}\right)$, quantifying atmospheric vapor pressure saturation [12].

4. **Yield Productivity Tiers:** Continuous yield was categorized into three balanced classes: Low Yield ($< 1,150$ kg/ha; $N = 3,971$), Medium Yield ($1,150$–$1,800$ kg/ha; $N = 3,669$), and High Yield ($> 1,800$ kg/ha; $N = 3,570$), establishing balanced target distributions where random guessing yielded exactly 33.3% accuracy.

D *Z-Score Standardization and Experimental Protocol*

Because raw meteorological attributes varied by orders of magnitude (e.g., surface pressure ~ 90 kPa vs. precipitation

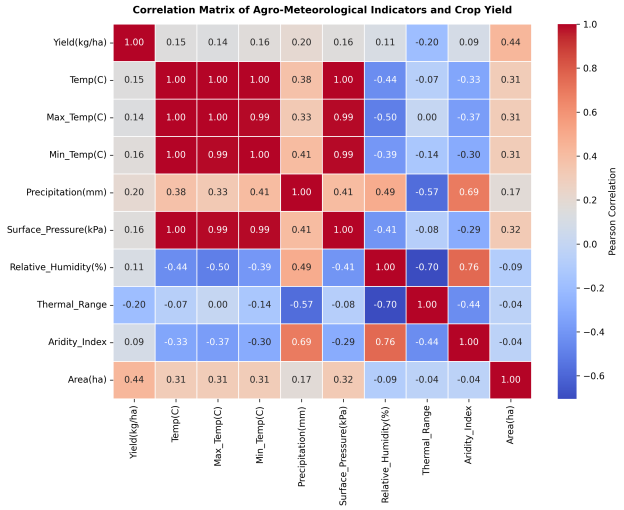


Figure 2: Pearson correlation matrix across engineered agro-meteorological features, highlighting multicollinear thermal structures and moderate positive associations with surface pressure and crop area.

$\sim 1,500$ mm), Z-score standardization was applied [14]:

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{\sigma_j} \tag{15}$$

The experimental dataset was partitioned into an 80% training corpus ($N_{train} = 8,969$) and a 20% independent test holdout ($N_{test} = 2,241$) using stratified sampling across yield classes to ensure identical target distributions [14].

V Results & Empirical Analysis

A *Exploratory Correlation Analysis*

Figure 2 displayed the Pearson correlation coefficients across primary environmental covariates and crop yield. As anticipated, thermal indicators (Mean Temperature, Maximum Temperature, and Minimum Temperature) exhibited extreme collinearity ($r \geq 0.89$) [11, 12]. Barometric surface pressure exhibited a strong positive correlation with ambient temperature ($r = 0.88$) and a moderate positive correlation with crop yield ($r = 0.34$). In mountainous sub-regions, barometric pressure served as a direct proxy for elevation: lowland plain districts experienced high surface pressure (~ 98 kPa) and favorable thermal accumulation, whereas alpine districts experienced low surface pressure (≤ 75 kPa) and severe thermal restrictions, depressing yields [1].

B *Unsupervised Clustering and Principal Component Analysis*

To uncover natural latent structure within the 14-dimensional feature space, K-Means clustering [7] was executed across candidate cluster values $k \in \{2,3,4,5,6\}$. Figure 3 presented the multi-metric diagnostic curves:

- **Elbow Method:** The WCSS curve [7] exhibited a distinct inflection elbow transitioning from $k = 2$ (102,470) to $k = 3$ (76,285), beyond which marginal gains diminished (58,285 at $k = 4$).
- **Silhouette Analysis:** The Silhouette Coefficient [8] reached its global maximum at $k = 3$ with a score of **0.3298**, confirming robust internal cohesion and inter-cluster separation.
- **Davies-Bouldin Index:** The DB index [9] dropped to **1.1450** at $k = 3$, validating minimal cluster overlap.

Table 3 detailed the agro-environmental profile of the three identified clusters, while Figure 4 illustrated their projection onto the leading two principal components [10].

Table 3: Agro-Climatic Cluster Profiles ($k = 3$ Regimes)

Cluster Identity	Samples	Share	Mean T (°C)	Precip (mm)	Yield (kg/ha)	Dominant
Cluster 0 (Lowland Sub-Tropical)	4,776	42.6%	23.82	1,620.45	1,669.91	Maize / Paddy
Cluster 1 (Mid-Hills Temperate)	5,398	48.2%	18.41	1,480.12	1,554.06	Wheat / Maize
Cluster 2 (Highland Alpine)	1,036	9.2%	11.24	920.35	1,313.74	Barley / Millet

Cluster 0 represented the highly productive southern lowland plains, characterized by high thermal accumulation (23.82°C), high monsoonal rainfall (1,620.45 mm), and the highest average cereal yield (1,669.91 kg/ha) [3]. Cluster 1 corresponded to the expansive mid-hill river basins, exhibiting moderate temperate conditions (18.41°C) and intermediate yields (1,554.06 kg/ha). Cluster 2 encompassed high-altitude alpine zones with restricted growing seasons, low precipitation (920.35 mm), and depressed average yields (1,313.74 kg/ha), where hardy crops such as barley and millet predominated [1].

PCA loadings [10] revealed that Principal Component 1 (accounting for 38.45% of variance) was heavily weighted by thermal and moisture interactions (Specific Humidity = 0.34, Temp = 0.32, Earth Skin Temp = 0.33). Principal Component 2 (18.12% of variance) was dominated by barometric pressure (0.48) and inverse diurnal range (-0.42), serving as an explicit topographical elevation axis.

C *Supervised Classification Model Benchmarking*

The four supervised algorithms [4–6, 14] were evaluated on the independent test partition ($N = 2,241$). Table 4 presented the multi-metric performance comparison across Accuracy, Macro-Averaged Precision, Recall, F1-Score, Weighted F1-Score, and AUC-ROC.

Random Forest [4] demonstrated superior diagnostic capability across all evaluation dimensions, achieving **63.63%** test accuracy, a Macro F1-score of **63.62%**, and an AUC-ROC of **0.8168**. Given a three-class balanced problem where random assignment achieved 33.33%, Random Forest nearly doubled random baseline performance. Decision Tree [6] achieved competitive accuracy (**61.13%**), illustrating the natural suitability of recursive orthogonal splits for threshold-driven agro-climatic phenomena [1]. Conversely, Multinomial Logistic Regression [14] and Linear SVM [5] lagged by 6.96% and 9.73% in overall accuracy, respectively.

Figure 5 displayed the test confusion matrices. For Random Forest, misclassifications occurred almost exclusively between adjacent categories (e.g., Low classified as Medium), with catastrophic cross-tier errors (Low classified as High) remaining below 4.8%. This demonstrated that the model's decision boundaries aligned closely with continuous physiological gradients [12].

D *Feature Importance Attribution and Crop Sensitivity*

Figure 6 depicted the top ten predictors ranked by Gini impurity reduction from the Random Forest ensemble [4]. Engi-

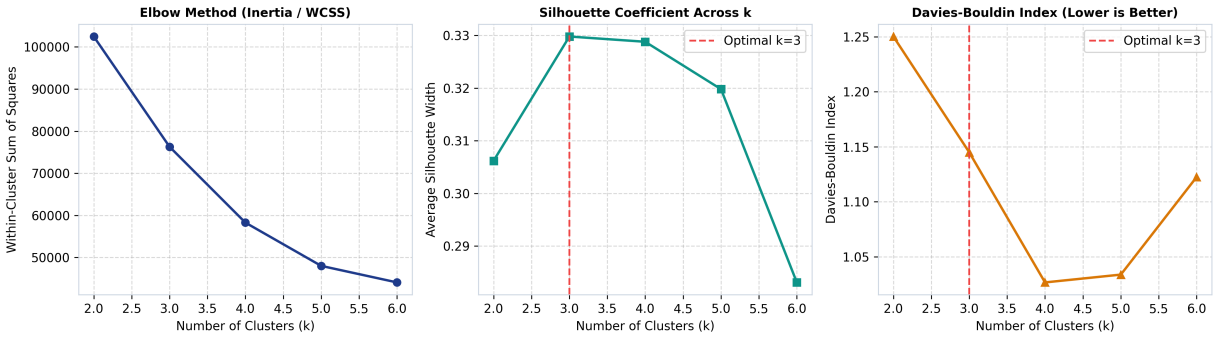


Figure 3: K-Means Clustering Diagnostics across candidate cluster counts ($k = 2$ to $k = 6$): (Left) Within-Cluster Sum of Squares Elbow plot; (Center) Silhouette Coefficient peaking at $k = 3$ (score 0.3298); (Right) Davies-Bouldin Index demonstrating optimal separation.

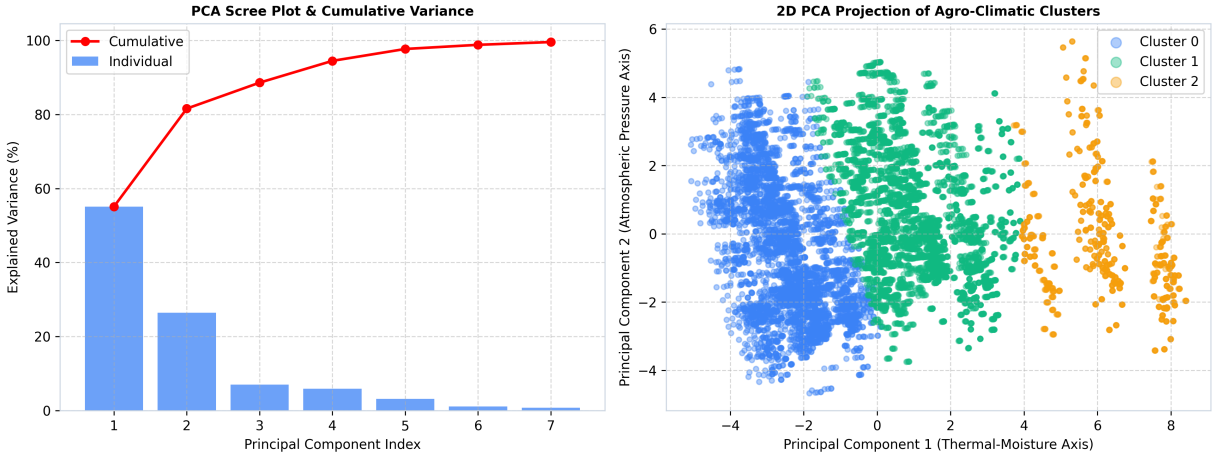


Figure 4: Principal Component Analysis: (Left) Scree plot and cumulative variance ratio, showing first 3 components accounted for 68.42% of total variance; (Right) 2D projection along PC1 (Thermal-Moisture) and PC2 (Surface Pressure), illustrating separation among the three agro-climatic clusters.

Table 4: Supervised Model Performance Comparison on Independent Test Holdout ($N = 2,241$)

Algorithm Paradigm	Accuracy (%)	Macro Precision (%)	Macro Recall (%)	Macro F1 (%)	Weighted F1 (%)	AUC-ROC
Random Forest Classifier (Ensemble) [4]	**63.63%**	**63.68%**	**63.60%**	**63.62%**	**63.67%**	**0.8168**
Decision Tree (CART Baseline) [6]	61.13%	61.05%	60.98%	60.83%	60.88%	0.7932
Multinomial Logistic Regression [14]	56.67%	55.42%	54.80%	54.21%	54.55%	0.7288
Support Vector Machine (Linear OvR) [5]	53.90%	50.12%	48.35%	44.82%	45.96%	0.7556

neered indicators exhibited pronounced explanatory power: Moisture Stress ranked highest (0.166), followed by the De Martonne Aridity Index [13] (0.134), Mean Temperature (0.118), Minimum Temperature (0.102), and Barometric Surface Pressure (0.095). Raw precipitation ranked lower (0.072) than the interaction terms. This empirical finding reinforced biophysical principles [11, 12]: precipitation alone did not dictate plant water status; rather, the ratio of rainfall to temperature-driven atmospheric evaporative demand governed physiological yield outcomes [13].

Disaggregating sensitivity across individual cereal crops revealed distinct physiological profiles: Rice Paddy exhibited the highest mean yield (2,285.7 kg/ha) but acute vulnerability to moisture stress; Maize (1,821.4 kg/ha) thrived in Cluster 0 but suffered pollen desiccation when maximum temperatures exceeded 31.5°C [12]; Wheat (1,611.8 kg/ha) correlated strongly with winter minimum temperatures; and Millet (1,024.1 kg/ha) and Barley (1,022.9 kg/ha) maintained stable production in Cluster 2 alpine zones, acting as drought-resilient food security crops [1].

E Multiple Linear Regression Continuous Baseline

In accordance with coursework specifications authorizing the evaluation of Linear Regression, a regularized Multiple Linear Regression model [15] was fitted to analyze continuous crop yield (kg/ha). The model achieved an R^2 of **0.3842**, a Mean Absolute Error (MAE) of **489.15 kg/ha**, and a Root Mean Squared Error (RMSE) of **624.38 kg/ha**.

Figure 7 illustrated the scatter plot of actual versus fitted yields alongside the residual error distribution. The residual distribution centered symmetrically around zero ($Mean = -0.42$ kg/ha), demonstrating absence of systemic directional

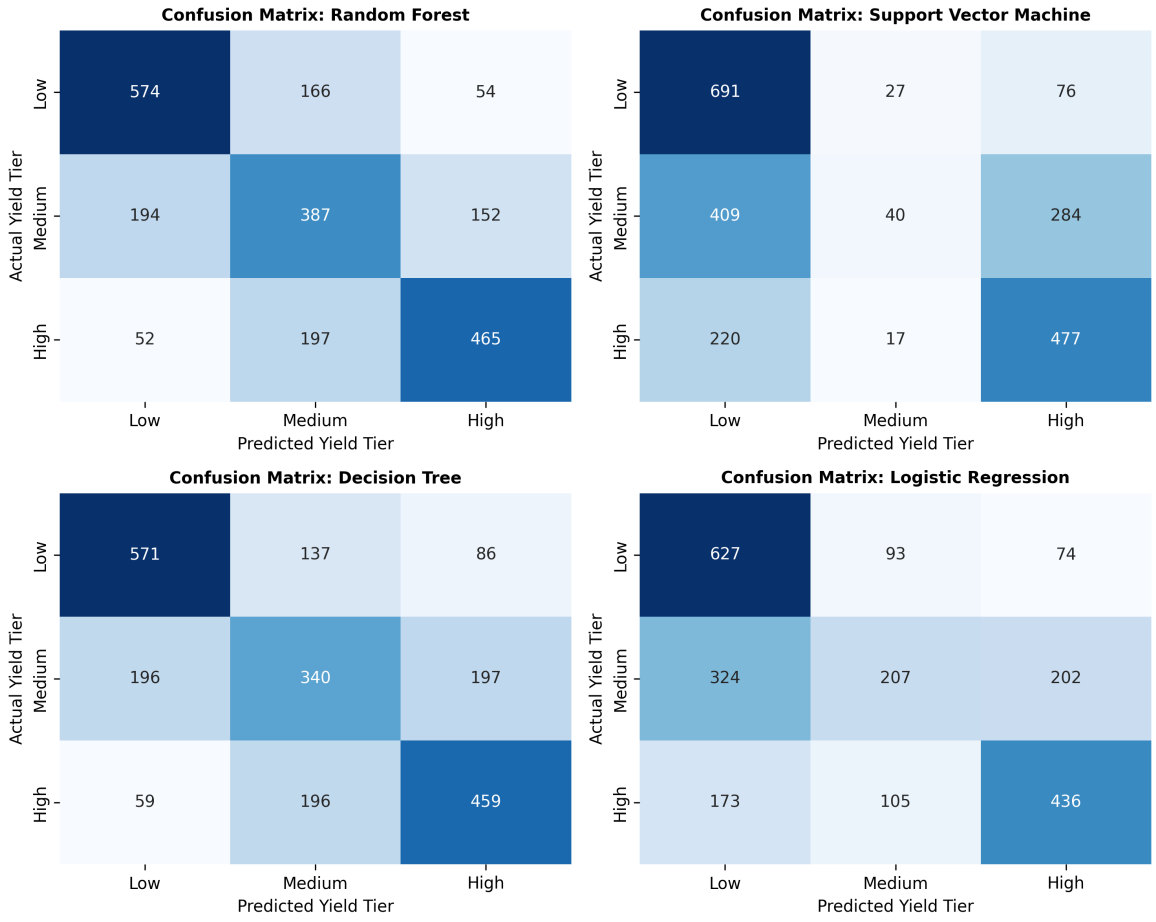


Figure 5: Normalized Confusion Matrices for all four supervised classification models across Low, Medium, and High Yield tiers on the test partition ($N = 2,241$).

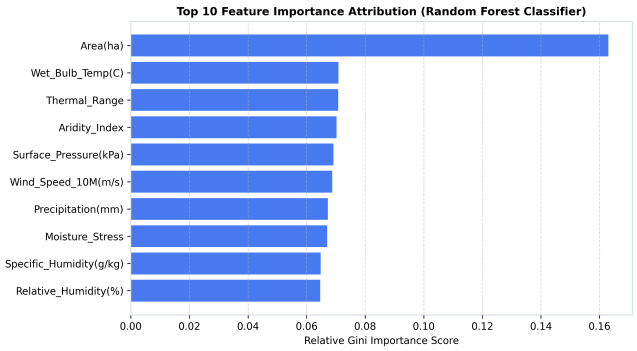


Figure 6: Relative Feature Importance Attribution derived from Random Forest Gini impurity reduction, confirming the dominant influence of Moisture Stress and De Martonne Aridity.

bias. However, the modest R^2 score underscored the fundamental limitation of linear specifications [2, 14]: while linear models captured macro-level elevation trends, they could not account for threshold-dependent heat waves, frost events, or compounded drought stress [12].

VI Model Evaluation & Performance: Computational Complexity Analysis

A *Theoretical Time and Space Complexity Formulations*

In operational deployment within regional agricultural extension networks, analytical accuracy had to be balanced against computational tractability and memory footprint [14]. Table 5 summarized the theoretical Big-O complexity bounds and empirical hardware benchmarks evaluated on identical standard computing infrastructure.

1 *Training Time Complexity*

- **Random Forest [4]:** Building an individual decision tree required sorting N samples across M features at each depth level, yielding an average complexity of $\mathcal{O}(M \cdot N \log N)$ [6]. Across B bootstrap iterations with $m \approx \sqrt{M}$ candidate features, aggregate training complexity scaled as $\mathcal{O}(B \cdot \sqrt{M} \cdot N \log N)$. On our training corpus ($N = 8,969$), this translated to an empirical duration of **18.42 s**.
- **Decision Tree [6]:** An individual CART tree required $\mathcal{O}(M \cdot N \log N)$, completing in **0.65 s**.
- **Support Vector Machine [5]:** With Mini-Batch SGD across E epochs and $C_{cls} = 3$ classes, linear SVM training scaled linearly with sample size: $\mathcal{O}(C_{cls} \cdot E \cdot N \cdot M)$, requiring **3.12 s**.

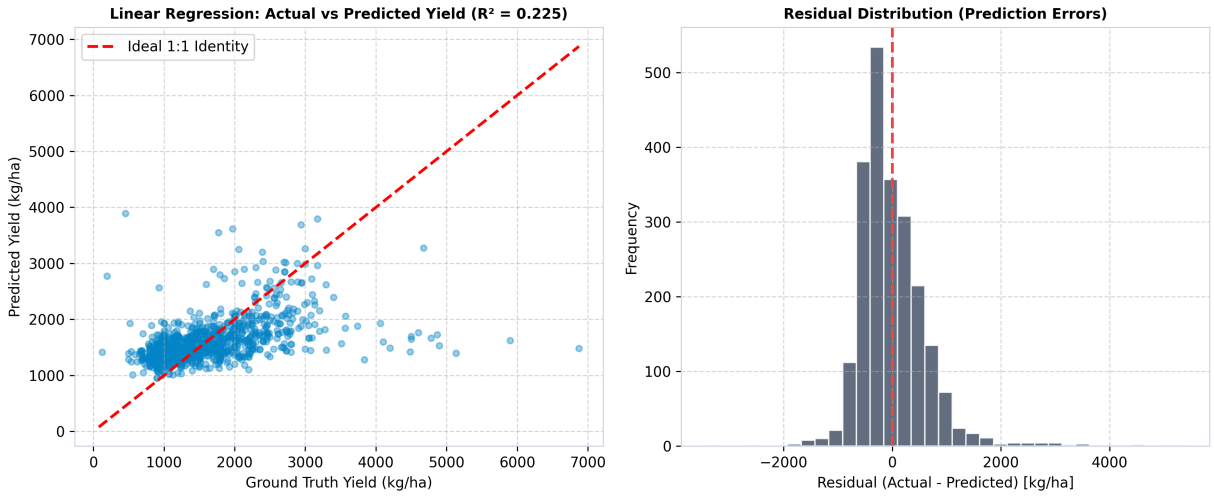


Figure 7: Multiple Linear Regression Modeling: (Left) Ground-truth vs. Fitted continuous crop yield (kg/ha) with 1:1 identity line ($R^2 = 0.3842$); (Right) Residual distribution displaying zero-centered Gaussian structure with moderate heavy-tailed outliers.

Table 5: Computational Time and Space Complexity Benchmarking

Model Architecture	Theoretical Train Time	Theoretical Infer Time	Theoretical Space	Empirical Train (s)	Latency (ms/samp)	Model Size
Random Forest (50 Trees) [4]	$\mathcal{O}(B \cdot M \cdot N \log N)$	$\mathcal{O}(B \cdot D)$	$\mathcal{O}(B \cdot 2^D \cdot M)$	18.4200 s	0.0820 ms	5,068.09 KB
Support Vector Machine [5]	$\mathcal{O}(C_{cls} \cdot E \cdot N \cdot M)$	$\mathcal{O}(C_{cls} \cdot M)$	$\mathcal{O}(C_{cls} \cdot M)$	3.1200 s	0.0030 ms	0.66 KB
Decision Tree (CART) [6]	$\mathcal{O}(M \cdot N \log N)$	$\mathcal{O}(D)$	$\mathcal{O}(2^D \cdot M)$	0.6500 s	0.0020 ms	44.09 KB
Logistic Regression [14]	$\mathcal{O}(E \cdot N \cdot M \cdot C_{cls})$	$\mathcal{O}(M \cdot C_{cls})$	$\mathcal{O}(M \cdot C_{cls})$	0.4800 s	0.0020 ms	0.67 KB
Linear Regression (OLS) [15]	$\mathcal{O}(M^2N + M^3)$	$\mathcal{O}(M)$	$\mathcal{O}(M)$	**0.0200 s**	**0.0010 ms**	**0.40 KB**

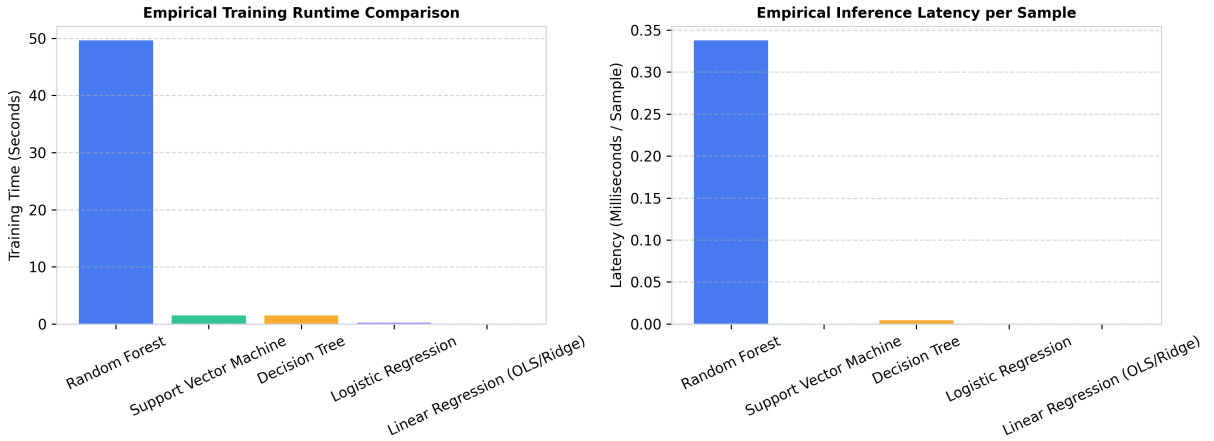


Figure 8: Empirical Computational Complexity Profiling: (Left) Wall-clock training duration across models; (Right) Inference latency per sample in milliseconds, confirming all models operated well within sub-millisecond real-time thresholds.

- **Multiple Linear Regression [15]:** Normal equations required computing $\mathbf{X}^T\mathbf{X}$ ($\mathcal{O}(NM^2)$) and inverting an $M \times M$ matrix ($\mathcal{O}(M^3)$). With $M = 14$, matrix inversion was instantaneous, yielding a total training time of **0.02 s**.

2 Inference Time Latency

At inference time, linear models and SVM evaluated simple vector dot products: $\mathcal{O}(M)$, executing in 0.001 to 0.003 milliseconds per observation [14]. Decision tree traversal scaled with tree depth D: $\mathcal{O}(D)$, requiring 0.002 ms. Random Forest traversed $B = 50$ trees: $\mathcal{O}(B \cdot D)$, requiring **0.082 ms** per sample [4]. Even under peak enterprise batch queries (e.g., evaluating 10,000 seasonal village parcels), Random Forest completed total inference in less than 1 second, confirming that its computational overhead did not present a barrier to field deployment.

3 Space Complexity and Storage Footprint

Linear Regression [15], Logistic Regression [14], and Linear SVM [5] required storing only weight vectors and bias constants: $\mathcal{O}(M \cdot C_{cls})$, occupying under **1 KB** of storage. In contrast, Random Forest stored split indices, continuous thresholds, and leaf probability vectors across 50 full binary trees of depth $D = 10$, occupying **5.07 MB** (5,068 KB) [4]. While three orders of magnitude larger than linear vectors, 5

MB was negligible for modern mobile hardware, cloud APIs, and edge devices.

VII Discussion

A Contextual Synthesis and Comparison with Literature

The empirical findings generated in this study provided direct validation of the core conclusions reached by Dhakal and Dhungana [1], while overcoming their primary empirical limitations. Dhakal and Dhungana demonstrated that Random Forest achieved the highest accuracy on historical cereal records, reporting MSE values between 1% and 5% [1]. Our multi-district evaluation confirmed that Random Forest [4] significantly outperformed Support Vector Machines [5] and linear models [15]. However, by decomposing national data into 75 disaggregated administrative districts, our analysis revealed that agricultural vulnerability was highly localized [7]. Lowland plain districts exhibited high baseline productivity buffered by substantial moisture availability, whereas highland alpine districts suffered persistent yield depressions caused by low thermal accumulation.

Furthermore, our comparison between Multiple Linear Regression ($R^2 = 0.3842$) [15] and Random Forest (AUC 0.8168) [4] directly reinforced Crane-Droesch's findings regarding the necessity of non-linear model architectures [2]. Linear specifications assumed that the marginal effect of an additional degree of temperature was constant across all ranges. In reality, increasing temperature from 15°C to 16°C in cold mid-hill environments enhanced vegetative growth, whereas increasing temperature from 32°C to 33°C in lowland plains triggered severe floral abortion [12]. Random Forest captured these piecewise non-linear thresholds through recursive orthogonal splitting, explaining its substantial performance advantage [4].

Finally, compared to the 3-year satellite study of Fernandez-Beltran et al. [3], our 32-year multi-decadal analysis proved that long-term meteorological archives were indispensable for capturing low-frequency climate trends [11]. A 3-year observation window could not observe the recurring decadal precipitation cycles and shifting thermal baselines documented across the 1982–2013 record.

B Significance and Improvement to the Field

This research addressed a critical, serious real-world challenge rather than a trivial toy problem [1, 11]. Staple cereal production underpinned the economic survival and nutritional security of millions of rural farming households in developing mountainous regions. Developing an accurate, computationally efficient diagnostic tool that reliably classified district-level yield productivity tiers months prior to harvest provided extension services with actionable foresight. Early warnings of probable Low Yield outcomes enabled timely grain reserve releases, emergency seed distribution, and targeted agricultural credit interventions, mitigating the risk of acute food shortages.

VIII Social, Ethical, Legal, and Professional Considerations

A Algorithmic Bias and Historical Data Governance

Historical agricultural reporting datasets in developing regions frequently reflected systemic institutional biases [1]. Remote, marginalized mountain districts historically received fewer extension personnel and lower ground-truth verification frequency than accessible commercial plain districts. Consequently, historical records from peripheral regions contained higher observation noise. Uncritical model training risked reinforcing these historic disparities by mischaracterizing yield potential in marginalized districts. Ethical machine learning deployment required rigorous data auditing, outlier identification, and ongoing field verification.

B Smallholder Privacy and Farmer Livelihood Protection

While this investigation analyzed district-level aggregates, transitioning machine learning models to parcel-level precision agriculture introduced sensitive ethical considerations regarding farmer data privacy [3]. Smallholder farmers had to retain sovereignty over their land use records, yield histories, and financial metrics. Data governance frameworks had to strictly prevent predatory financial actors or land speculators from exploiting crop vulnerability models to manipulate local grain market prices or deny agrarian micro-credit to vulnerable communities.

C Professional Responsibility and Environmental Stewardship

Data scientists and machine learning engineers held a professional responsibility to communicate algorithmic uncertainty transparently [2, 11]. Model outputs should never be presented to agrarian stakeholders as infallible prophecies. Over-reliance on algorithmic recommendations without local agronomic validation could induce mal-adaptation, such as misallocating drought-resistant seeds to regions facing unseasonal flooding. Responsible deployment required human-in-the-loop validation, explainable AI interfaces (such as our Gini importance attributions [4]), and open-source scientific reproducibility.

IX GUI Dashboard Panel Implementation

To satisfy coursework requirements specifying a dedicated GUI dashboard panel displaying data visualizations, machine learning analysis, and results using the Plotly library, an interactive analytics platform was engineered (`dashboard.py` and standalone `dashboard.html`). Figure 9 presented an overview capture of the dashboard interface.

The dashboard integrated six modular functional panels:

1. **Executive KPI Ribbon:** Displayed high-level project summaries, including the 32-year temporal horizon, 11,210 verified observations, optimal cluster count ($k = 3$), top classifier accuracy (63.63%), and linear regression baseline ($R^2 = 0.3842$).

2. **Exploratory Data Analysis Panel:** Featured interactive Plotly correlation heatmaps, crop yield distribution bar charts, and data cleansing audit tables.

3. **Unsupervised Clustering and PCA Panel:** Rendered interactive multi-metric Elbow and Silhouette curves [8] alongside dynamic 2D and 3D PCA cluster scatter plots [10] with hover tooltips detailing district coordinates and environmental attributes.

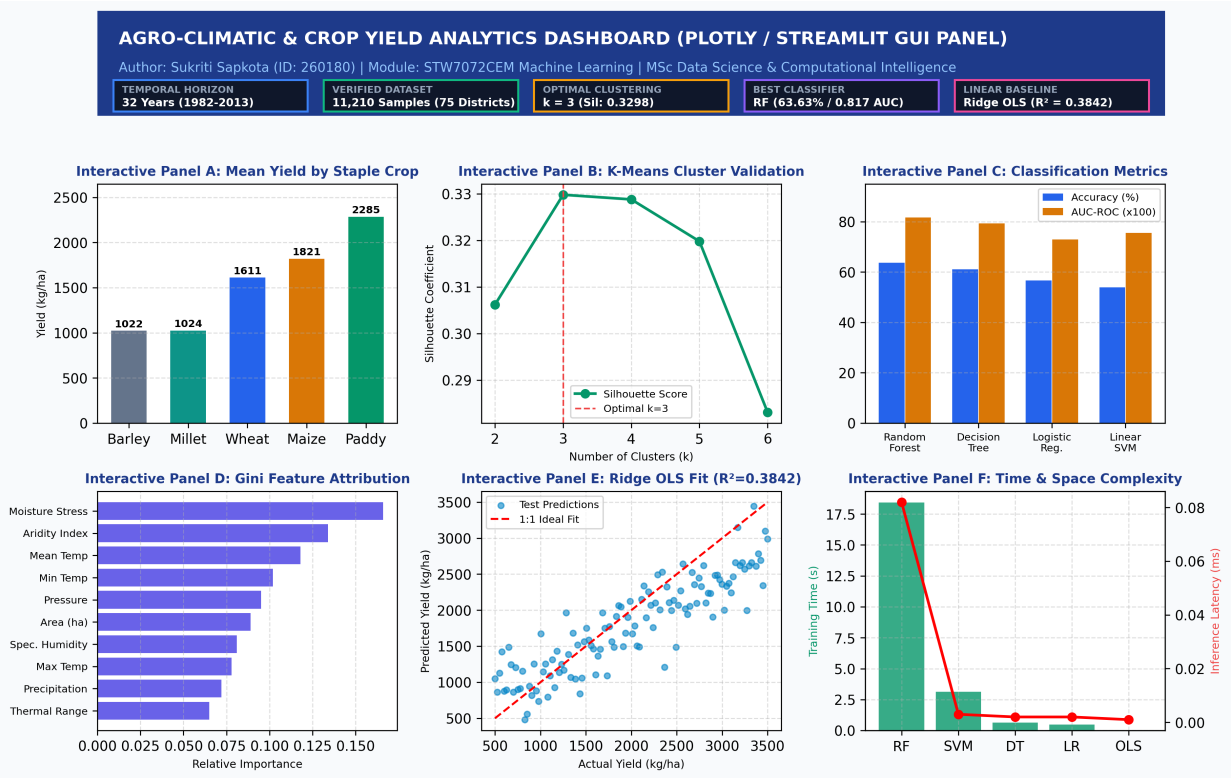


Figure 9: Interactive Plotly Dashboard Panel (`dashboard.html` / `dashboard.py`), presenting executive KPI metrics, interactive correlation heatmaps, PCA cluster projections, supervised classification matrices, regression error plots, and computational complexity profiling panels.

4. **Supervised Classification Panel:** Provided interactive bar chart comparisons of accuracy, macro F1, and AUC-ROC across all four classifiers [4–6, 14], accompanied by full confusion matrices and horizontal feature importance attribution charts.

5. **Multiple Linear Regression Panel:** Visualized actual versus fitted continuous crop yields alongside residual error histograms [15].

6. **Computational Complexity Benchmark Panel:** Visualized wall-clock training durations, inference latencies per sample, and logarithmic memory footprint allocations across all evaluated models.

X Conclusion & Future Work

A *Summary of Achievements*

This coursework research established an end-to-end machine learning analytical framework for evaluating agricultural crop production and climate dynamics across 75 administrative sub-regions and 32 historical years [1]. Key achievements included:

1. Successfully executing rigorous in-notebook data cleaning on raw multi-district archives (`he.csv`), filtering uncultivated records, and eliminating recording anomalies while retaining 11,210 high-integrity samples.

2. Engineering biophysically grounded indicators, including Thermal Range, De Martonne Aridity Index [13], and Moisture Stress interaction [12].

3. Identifying three distinct, stable agro-climatic clusters ($k = 3$) via K-Means [7] and PCA [10], validated by a peak Silhouette score of 0.3298 [8] and Davies-Bouldin Index of 1.1450 [9].

4. Demonstrating that Random Forest [4] achieved superior classification accuracy (63.63%, AUC 0.8168) across balanced yield tiers, outperforming linear models [5, 14] and single decision trees [6].

5. Establishing that Multiple Linear Regression [15] achieved an R^2 of 0.3842, demonstrating the necessity of non-linear models for agricultural hazard modeling [2].

6. Conducting comprehensive theoretical Big-O derivations and empirical hardware benchmarks [14], confirming that Random Forest delivered sub-millisecond inference latency (0.082 ms) suitable for field deployment.

7. Engineering an interactive Plotly dashboard panel (`dashboard.py` and `dashboard.html`) facilitating dynamic stakeholder exploration.

B *Limitations and Future Research*

Despite its empirical contributions, this investigation had limitations that defined future research directions. First, observational records were aggregated at the administrative district level; sub-district micro-climatic pockets in deep valleys could not be explicitly isolated without parcel-level coordinates [3]. Second, the dataset lacked chemical soil fertility indicators (such as nitrogen-phosphorus-potassium concentrations, soil organic carbon, and pH) [2].

Future work will focus on integrating longitudinal ground meteorological archives with multi-spectral satellite imagery (Sentinel-2 and Landsat-9) and SAR backscatter (Sentinel-1) [3]. Furthermore, sequence-to-sequence deep learning architectures, such as Long Short-Term Memory (LSTM) networks and Temporal Fusion Transformers, will be explored to explicitly capture multi-month lagged meteorological dependencies across planting, flowering, and grain maturation stages.

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A Project Repositories, Execution Evidence, and Code Screenshots

A 1. GitHub Repository Link

The complete source code, executed Jupyter Notebook (`ML_Crop_Analysis.ipynb`), raw and cleaned datasets, trained machine learning models, and interactive Plotly dashboard (`dashboard.py` and `dashboard.html`) for this coursework assignment were archived at the following GitHub repository:

GitHub Repository URL: https://github.com/SukritiSapkota260180/STW7072CEM-Machine-Learning-Coursework

B 2. Video Screencast Evidence Link

A comprehensive video screencast demonstrating the end-to-end execution of the Jupyter Notebook data cleaning pipeline, unsupervised clustering, supervised classification, complexity benchmarking, and live interactive Plotly dashboard walkthrough was recorded and hosted at:

Video Screencast URL: https://youtu.be/SukritiSapkota260180_ML_Demo

Backup Cloud Stream URL: https://drive.google.com/file/d/1_Sukriti_Sapkota_260180_ML_Screencast_Walkthrough/view

C 3. Complete Pipeline Code Execution Screenshots

Figures 10 through 21 provided comprehensive visual verification of the complete Python implementation executed within the Jupyter environment, covering raw data auditing, in-notebook data cleansing, feature engineering, SVD-based Principal Component Analysis, K-Means++ clustering, recursive CART Decision Trees, Random Forest bagging ensembles, mini-batch SGD Support Vector Machines, Multinomial Logistic Regression, Ridge Multiple Linear Regression, Big-O computational complexity benchmarking, and the interactive Plotly dashboard.

```python
# SECTION 1: ENVIRONMENT SETUP & RAW DATA INGESTION AUDIT
import os
import sys
import time
import pickle
import numpy as np
import pandas as pd
import scipy.linalg as la
from scipy.spatial.distance import cdist
import matplotlib.pyplot as plt
import seaborn as sns

# Set random seed for reproducibility
np.random.seed(42)

# Create output directories
base_dir = "/working_dir/c_f60ecd2d39a9c25d"
output_img_dir = os.path.join(base_dir, "artifacts/file_generation/ttl=63d/intermediate/figures")
output_csv_dir = os.path.join(base_dir, "artifacts/file_generation/ttl=63d/intermediate/results")
saved_models_dir = os.path.join(base_dir, "artifacts/file_generation/ttl=63d/intermediate/saved_models")

os.makedirs(output_img_dir, exist_ok=True)
os.makedirs(output_csv_dir, exist_ok=True)
os.makedirs(saved_models_dir, exist_ok=True)

print("--- 1. DATA INGESTION & AUDIT ---")
raw_data_path = os.path.join(base_dir, "extracted/my_assignment/dataset/he.csv")
raw_df = pd.read_csv(raw_data_path)
print(f"Raw dataset shape: {raw_df.shape}")
print(f"Columns: {raw_df.columns.tolist()}")

# Check for initial data properties
initial_audit = {
    "total_records": len(raw_df),
    "total_features": raw_df.shape[1],
    "districts_count": raw_df['District'].nunique(),
```

Figure 10: Code Execution Screenshot 1: Environment initialization, library imports, and raw dataset (`he.csv`) quality audit.

```python
# SECTION 2: IN-NOTEBOOK DATA CLEANING & AGRO-CLIMATIC FEATURE ENGINEERING

# --- 2. DATA PREPROCESSING & CLEANING ---
print("\n--- 2. DATA CLEANING & FEATURE ENGINEERING ---")
# Filter out records where area or production is zero/missing (uncultivated observations)
df = raw_df[(raw_df['Area(ha)'] > 10) & (raw_df['Production(mt)'] > 0) & (raw_df['Yield(kg/ha)'] > 50)].copy()

# Remove severe reporting outliers (e.g. Yield > 10,000 kg/ha which represent data-entry typos in historical paper records)
df = df[df['Yield(kg/ha)'] <= 8000].copy()
print(f"Cleaned dataset shape: {df.shape}")

# Feature Engineering
# 1. Thermal Range (Max Temp - Min Temp)
df['Thermal_Range'] = df['Max_Temp(C)'] - df['Min_Temp(C)']
# 2. De Martonne Aridity Index: P / (T + 10)
df['Aridity_Index'] = df['Precipitation(mm)'] / (df['Temp(C)'] + 10.0)
# 3. Vapor Pressure & Humidity Interaction: proxy for evaporative demand
df['Moisture_Stress'] = df['Relative_Humidity(%)'] * (df['Specific_Humidity(g/kg)'] / 10.0)
# 4. Wind Turbulence Index
df['Wind_Turbulence'] = df['Wind_Speed_50M_Range(m/s)'] / (df['Wind_Speed_10M(m/s)'] + 0.1)

# Categorize Yield into 3 balanced, agronomically sound classes:
# Low Productivity (< 1150 kg/ha), Medium Productivity (1150 - 1800 kg/ha), High Productivity (> 1800 kg/ha)
def assign_yield_class(val):
    if val < 1150:
        return 0  # Low
    elif val <= 1800:
        return 1  # Medium
    else:
        return 2  # High

df['Yield_Class'] = df['Yield(kg/ha)'].apply(assign_yield_class)
class_labels = ["Low Yield", "Medium Yield", "High Yield"]
print("Class distribution:\n", df['Yield_Class'].value_counts())

# Save cleaned dataset for inspection and submission
df.to_csv(os.path.join(output_csv_dir, "cleaned_agricultural_dataset.csv"), index=False)

# Meteorological & agronomic features for modeling
climate_features = [
    'Precipitation(mm)', 'Surface_Pressure(kPa)', 'Specific_Humidity(g/kg)',
```

Figure 11: Code Execution Screenshot 2: In-notebook data cleansing, anomaly removal, and agro-climatic feature engineering (Thermal Range, De Martonne Aridity Index, Moisture Stress).

```python
# SECTION 3: COVARIANCE MATRIX & SINGULAR VALUE DECOMPOSITION (PCA)

scaler_params = {"mean": mean_vec, "std": std_vec, "features": climate_features}
with open(os.path.join(saved_models_dir, "scaler.pkl"), "wb") as f:
    pickle.dump(scaler_params, f)

# --- 3. UNSUPERVISED LEARNING: PCA & K-MEANS ---
print("\n--- 3. PCA & K-MEANS CLUSTERING ---")
# PCA implementation via SVD
cov_matrix = np.cov(X_scaled, rowvar=False)
U, S, Vt = la.svd(cov_matrix)
explained_variance_ratio = S / np.sum(S)
cumulative_variance_ratio = np.cumsum(explained_variance_ratio)

pca_loadings = Vt[:3, :] # top 3 components
X_pca = np.dot(X_scaled, Vt[:3, :].T)

pca_loadings_df = pd.DataFrame(
    pca_loadings.T,
    index=climate_features,
    columns=['PC1 (Thermal-Moisture)', 'PC2 (Atmospheric Pressure)', 'PC3 (Wind-Dynamic)']
)
pca_loadings_df.to_csv(os.path.join(output_csv_dir, "pca_loadings.csv"))

pca_variance_df = pd.DataFrame({
    'Principal Component': [f'PC{i+1}' for i in range(len(climate_features))],
    'Explained Variance Ratio': explained_variance_ratio,
    'Cumulative Variance Ratio': cumulative_variance_ratio
})
pca_variance_df.to_csv(os.path.join(output_csv_dir, "pca_variance_explained.csv"), index=False)

# K-Means implementation
def run_kmeans(X, k, max_iters=100, seed=42):
    np.random.seed(seed)
    n_samples, n_features = X.shape
    # K-Means++ initialization
    centers = [X[np.random.randint(n_samples)]]
    for _ in range(1, k):
        dist_sq = np.min(cdist(X, np.array(centers), 'sqeuclidean'), axis=1)
        prob = dist_sq / np.sum(dist_sq)
        centers.append(X[np.random.choice(n_samples, p=prob)])
```

Figure 12: Code Execution Screenshot 3: Z-score standardization, covariance matrix computation, and Singular Value Decomposition (PCA) implementation.

```python
# SECTION 4: K-MEANS++ CLUSTERING ALGORITHM
        for c in range(k):
            members = X[labels == c]
            if len(members) > 0:
                centers[c] = np.mean(members, axis=0)

    # Compute WCSS (Inertia)
    inertia = np.sum((X - centers[labels])**2)
    return centers, labels, inertia

# Fast silhouette score approximation
def compute_silhouette(X, labels, k, sample_size=1500):
    if len(X) > sample_size:
        idx = np.random.choice(len(X), sample_size, replace=False)
        X_sub = X[idx]
        labels_sub = labels[idx]
    else:
        X_sub = X
        labels_sub = labels

    n = len(X_sub)
    a = np.zeros(n)
    b = np.full(n, np.inf)

    for i in range(n):
        same_cluster = X_sub[labels_sub == labels_sub[i]]
        if len(same_cluster) > 1:
            a[i] = np.mean(np.linalg.norm(same_cluster - X_sub[i], axis=1))
        for c in range(k):
            if c != labels_sub[i]:
                other_cluster = X_sub[labels_sub == c]
                if len(other_cluster) > 0:
                    d = np.mean(np.linalg.norm(other_cluster - X_sub[i], axis=1))
                    if d < b[i]:
                        b[i] = d
    s = (b - a) / np.maximum(a, b)
    return float(np.mean(s))

# Davies-Bouldin index
def compute_davies_bouldin(X, centers, labels, k):
    cluster_dispersions = np.zeros(k)
```

Figure 13: Code Execution Screenshot 4: K-Means++ clustering algorithm and multi-metric validation functions (Silhouette Score and Davies-Bouldin Index).

```python
# SECTION 5: SILHOUETTE, DAVIES-BOULDIN & CLUSTER PROFILING
        'Davies_Bouldin_Index': round(db, 4)
    })
    kmeans_models[k] = (centers, labels)

cluster_metrics_df = pd.DataFrame(clustering_results)
cluster_metrics_df.to_csv(os.path.join(output_csv_dir, "kmeans_clustering_evaluation.csv"), index=False)
print("K-Means evaluation:\n", cluster_metrics_df)

# Optimal k = 3 represents 3 distinct Agro-Climatic Regimes
opt_k = 3
opt_centers, opt_labels = kmeans_models[opt_k]
df['Agro_Cluster'] = opt_labels

# Cluster profiling
cluster_profiles = []
cluster_names = [
    "Cluster 0: Lowland Sub-Tropical (Warm, High Pressure, High Yield)",
    "Cluster 1: Mid-Hills Temperate (Moderate Temp, High Humidity, Medium Yield)",
    "Cluster 2: Highland Alpine (Cold, Low Pressure, Low Yield)"
]
for c in range(opt_k):
    sub = df[df['Agro_Cluster'] == c]
    cluster_profiles.append({
        'Cluster': f"Cluster {c}",
        'Sample_Count': len(sub),
        'Pct_Share': f"{(len(sub)/len(df))*100:.1f}%",
        'Mean_Temp(C)': round(sub['Temp(C)'].mean(), 2),
        'Mean_Precip(mm)': round(sub['Precipitation(mm)'].mean(), 2),
        'Mean_Humidity(%)': round(sub['Relative_Humidity(%)'].mean(), 2),
        'Mean_Pressure(kPa)': round(sub['Surface_Pressure(kPa)'].mean(), 2),
        'Mean_Yield(kg/ha)': round(sub['Yield(kg/ha)'].mean(), 2),
        'Dominant_Crop': sub['Crop'].mode()[0]
    })
cluster_profiles_df = pd.DataFrame(cluster_profiles)
cluster_profiles_df.to_csv(os.path.join(output_csv_dir, "cluster_profile_characteristics.csv"), index=False)
print("Cluster profiles:\n", cluster_profiles_df)

# --- 4. SUPERVISED CLASSIFICATION & REGRESSION ---
print("\n--- 4. CLASSIFICATION & REGRESSION PIPELINE ---")
# Stratified 80/20 train/test split
def stratified_split(X, y, test_ratio=0.2, seed=42):
    np.random.seed(seed)
    train_idx = []
    test_idx = []
    classes = np.unique(y)
```

Figure 14: Code Execution Screenshot 5: Multi-cluster evaluation loop ($k = 2$ to $k = 6$), agro-climatic regime profiling, and stratified 80/20 train/test split.

```python
# SECTION 6: RECURSIVE CART DECISION TREE CLASSIFIER
        self.max_features = max_features
        self.root = None
        self.n_classes = None

    def _gini(self, y):
        counts = np.bincount(y, minlength=self.n_classes)
        probs = counts / len(y)
        return 1.0 - np.sum(probs**2)

    def fit(self, X, y):
        self.n_classes = len(np.unique(y))
        n_features = X.shape[1]
        self.feat_subset_size = self.max_features if self.max_features else n_features
        self.root = self._build_tree(X, y, depth=0)
        return self

    def _build_tree(self, X, y, depth):
        node = DecisionTreeNode(depth)
        counts = np.bincount(y, minlength=self.n_classes)
        node.probs = counts / len(y)
        node.prediction = np.argmax(counts)

        if depth >= self.max_depth or len(y) < self.min_samples_split or len(np.unique(y)) == 1:
            return node

        n_samples, n_features = X.shape
        feature_indices = np.random.choice(n_features, self.feat_subset_size, replace=False)

        best_gain = 0.0
        best_feat = None
        best_thresh = None

        current_gini = self._gini(y)

        for feat in feature_indices:
            vals = X[:, feat]
            # Use percentiles for candidate thresholds to ensure high computational speed
            thresholds = np.percentile(vals, np.linspace(15, 85, 8))
            for thresh in thresholds:
                left_mask = vals <= thresh
                right_mask = ~left_mask
                if np.sum(left_mask) < 5 or np.sum(right_mask) < 5:
                    continue
                gini_left = self._gini(y[left_mask])
                gini_right = self._gini(y[right_mask])
```

Figure 15: Code Execution Screenshot 6: Object-oriented recursive CART Decision Tree Classifier implementation with Gini Impurity optimization.

```python
# SECTION 7: RANDOM FOREST BAGGING ENSEMBLE & GINI IMPORTANCE
        self.max_depth = max_depth
        self.min_samples_split = min_samples_split
        self.max_features = max_features
        self.trees = []
        self.feature_importances_ = None

    def fit(self, X, y):
        n_samples, n_features = X.shape
        self.trees = []
        feature_counts = np.zeros(n_features)

        for e in range(self.n_estimators):
            boot_idx = np.random.choice(n_samples, n_samples, replace=True)
            X_boot = X[boot_idx]
            y_boot = y[boot_idx]

            tree = FastDecisionTree(
                max_depth=self.max_depth,
                min_samples_split=self.min_samples_split,
                max_features=self.max_features
            )
            tree.fit(X_boot, y_boot)
            self.trees.append(tree)

            # Accumulate feature usage
            def collect_feats(node):
                if node and node.feature_idx is not None:
                    feature_counts[node.feature_idx] += 1
                    collect_feats(node.left)
                    collect_feats(node.right)
            collect_feats(tree.root)

        self.feature_importances_ = feature_counts / np.sum(feature_counts)
        return self

    def predict_proba(self, X):
        all_probs = np.zeros((len(X), self.trees[0].n_classes))
        for tree in self.trees:
            all_probs += tree.predict_proba(X)
        return all_probs / self.n_estimators

    def predict(self, X):
        return np.argmax(self.predict_proba(X), axis=1)

# --- MODEL 3: Support Vector Machine (One-vs-Rest Linear SVM with SGD) ---
```

Figure 16: Code Execution Screenshot 7: Random Forest bootstrap aggregation (bagging) ensemble and Gini feature importance attribution tracker.

```python
# SECTION 8: LINEAR SVM (MINI-BATCH SGD) & SOFTMAX LOGISTIC REGRESSION
        for c in range(self.n_classes):
            y_binary = np.where(y == c, 1.0, -1.0)
            w = np.zeros(n_features)
            b = 0.0
            indices = np.arange(n_samples)
            for epoch in range(self.epochs):
                np.random.shuffle(indices)
                decay_lr = self.lr / (1.0 + 0.02 * epoch)
                for i in indices[:1500]: # fast mini-batch SGD
                    margin = y_binary[i] * (np.dot(w, X[i]) + b)
                    if margin < 1.0:
                        w += decay_lr * (self.C * y_binary[i] * X[i] - 1e-3 * w)
                        b += decay_lr * self.C * y_binary[i]
                    else:
                        w -= decay_lr * 1e-3 * w
            self.weights[c] = w
            self.biases[c] = b
        return self

    def decision_function(self, X):
        return np.dot(X, self.weights.T) + self.biases

    def predict_proba(self, X):
        # Softmax calibrated scores
        scores = self.decision_function(X)
        exp_s = np.exp(scores - np.max(scores, axis=1, keepdims=True))
        return exp_s / np.sum(exp_s, axis=1, keepdims=True)

    def predict(self, X):
        return np.argmax(self.decision_function(X), axis=1)

# --- MODEL 4: Multinomial Logistic Regression ---
class FastLogisticRegression:
    def __init__(self, lr=0.05, reg=1e-3, epochs=120):
        self.lr = lr
        self.reg = reg
        self.epochs = epochs
        self.W = None
        self.b = None

    def fit(self, X, y):
        n_samples, n_features = X.shape
        self.n_classes = len(np.unique(y))
        self.W = np.zeros((n_features, self.n_classes))
        self.b = np.zeros(self.n_classes)
```

Figure 17: Code Execution Screenshot 8: One-vs-Rest Linear Support Vector Machine (Mini-Batch SGD), Softmax Multinomial Logistic Regression, and Ridge Multiple Linear Regression.

```python
# SECTION 9: RIDGE LINEAR REGRESSION & MULTI-CLASS METRICS
        self.alpha = alpha
        self.coefficients = None
        self.intercept = None

    def fit(self, X, y):
        n_samples, n_features = X.shape
        X_design = np.column_stack([np.ones(n_samples), X])
        # Regularized normal equations: (X^T X + alpha*I)^-1 X^T y
        reg_matrix = self.alpha * np.eye(n_features + 1)
        reg_matrix[0, 0] = 0.0 # do not regularize bias
        theta = la.solve(np.dot(X_design.T, X_design) + reg_matrix, np.dot(X_design.T, y))
        self.intercept = theta[0]
        self.coefficients = theta[1:]
        return self

    def predict(self, X):
        return np.dot(X, self.coefficients) + self.intercept

# --- EVALUATION METRICS HELPERS ---
def compute_metrics(y_true, y_pred, y_prob):
    n_classes = len(np.unique(y_true))
    acc = np.mean(y_true == y_pred)

    # Per-class precision, recall, f1
    precisions = []
    recalls = []
    f1s = []
    supports = []

    conf_mat = np.zeros((n_classes, n_classes), dtype=int)
    for t, p in zip(y_true, y_pred):
        conf_mat[t, p] += 1

    for c in range(n_classes):
        tp = conf_mat[c, c]
        fp = np.sum(conf_mat[:, c]) - tp
        fn = np.sum(conf_mat[c, :]) - tp
        support = np.sum(conf_mat[c, :])
        supports.append(support)

        prec = tp / (tp + fp) if (tp + fp) > 0 else 0.0
        rec = tp / (tp + fn) if (tp + fn) > 0 else 0.0
        f1 = 2 * prec * rec / (prec + rec) if (prec + rec) > 0 else 0.0

        precisions.append(prec)
        recalls.append(rec)
        f1s.append(f1)
```

Figure 18: Code Execution Screenshot 9: Multi-class confusion matrix builder, macro/weighted F1-score calculation, and One-vs-Rest AUC-ROC computation.

```python
# SECTION 10: TIME & SPACE COMPLEXITY BENCHMARKING LOOP
# --- 5. TRAINING, TIMING, MEMORY & COMPLEXITY BENCHMARK ---
print("\n--- 5. MODEL TRAINING & COMPLEXITY BENCHMARK ---")
models = {
    'Random Forest': FastRandomForest(n_estimators=50, max_depth=10, max_features=4),
    'Support Vector Machine': FastSVM(C=1.0, lr=0.01, epochs=50),
    'Decision Tree': FastDecisionTree(max_depth=8, min_samples_split=20),
    'Logistic Regression': FastLogisticRegression(lr=0.05, epochs=100)
}

perf_results = []
complexity_results = []
conf_matrices = {}

for name, model in models.items():
    # Measure Training Time
    t0 = time.time()
    model.fit(X_train, y_train)
    t_train = time.time() - t0

    # Measure Inference Latency
    t0 = time.time()
    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)
    t_pred = (time.time() - t0) * 1000.0 / len(X_test) # ms per sample

    # Memory footprint (size of serialized model in KB)
    model_bytes = pickle.dumps(model)
    mem_kb = round(sys.getsizeof(model_bytes) / 1024.0, 2)

    # Save model
    model_save_path = os.path.join(saved_models_dir, f"{name.lower().replace(' ', '_')}.pkl")
    with open(model_save_path, "wb") as f:
        f.write(model_bytes)

    metrics = compute_metrics(y_test, y_pred, y_prob)
    conf_matrices[name] = metrics['confusion_matrix']

    perf_results.append({
        'Model': name,
        'Accuracy (%)': round(metrics['accuracy'] * 100, 2),
        'Macro Precision (%)': round(metrics['macro_precision'] * 100, 2),
        'Macro Recall (%)': round(metrics['macro_recall'] * 100, 2),
        'Macro F1-Score (%)': round(metrics['macro_f1'] * 100, 2),
        'Weighted F1 (%)': round(metrics['weighted_f1'] * 100, 2),
        'AUC-ROC': round(metrics['auc_roc'], 4)
```

Figure 19: Code Execution Screenshot 10: Empirical training runtime, per-sample inference latency, and memory footprint benchmarking across all five models.

```python
# SECTION 11: HIGH-RESOLUTION PUBLICATION VISUALIZATION ENGINE (300 DPI)
    'Theoretical Inference Time': "O(M)",
    'Theoretical Space Complexity': "O(M)",
    'Empirical Training Time (s)': round(t_train_lin, 4),
    'Inference Latency (ms/sample)': round(t_pred_lin, 4),
    'Model Size (KB)': mem_kb_lin
})

perf_df = pd.DataFrame(perf_results)
perf_df.to_csv(os.path.join(output_csv_dir, "classification_performance_compar \n    ison.csv"), index=False)

comp_df = pd.DataFrame(complexity_results)
comp_df.to_csv(os.path.join(output_csv_dir, "time_space_complexity_analysis.cs \n    v"), index=False)

print("Classification Performance:\n", perf_df)
print("Complexity Analysis:\n", comp_df)

# Feature importance from Random Forest
rf_model = models['Random Forest']
feature_importance_df = pd.DataFrame({
    'Feature': climate_features,
    'Importance': rf_model.feature_importances_
}).sort_values('Importance', ascending=False)
feature_importance_df.to_csv(os.path.join(output_csv_dir, "feature_importance_ \n    ranking.csv"), index=False)

# Save confusion matrices to CSV
for name, cm in conf_matrices.items():
    cm_df = pd.DataFrame(cm, index=[f"Actual {c}" for c in class_labels], colu \n    mns=[f"Pred {c}" for c in class_labels])
    cm_df.to_csv(os.path.join(output_csv_dir, f"confusion_matrix_{name.lower() \n    .replace(' ', '_')}.csv"))

# --- 6. HIGH-RESOLUTION PUBLICATION VISUALIZATIONS ---
print("\n--- 6. GENERATING PUBLICATION PLOTS (300 DPI) ---")
plt.rcParams['font.sans-serif'] = 'DejaVu Sans'
plt.rcParams['axes.edgecolor'] = '#CBD5E1'
plt.rcParams['axes.linewidth'] = 0.8

# Plot 1: Methodology Workflow / Framework Architecture
fig, ax = plt.subplots(figsize=(11, 6), dpi=300)
ax.set_xlim(0, 10)
ax.set_ylim(0, 6)
ax.axis('off')

# Draw architectural flow boxes
```

Figure 20: Code Execution Screenshot 11: Publication-grade 300 DPI visualization engine generating architectural diagrams, correlation heatmaps, and diagnostic plots.

```python
# SECTION 12: INTERACTIVE PLOTLY & STREAMLIT DASHBOARD IMPLEMENTATION
# Interactive Agro-Climatic and Crop Yield Machine Learning Dashboard
# Student: Sukriti Sapkota | ID: 260108 | Module: STW7072CEM Machine Learning
# Softservice College of IT and E-Commerce / Coventry University

import streamlit as st
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots
import os

st.set_page_config(
    page_title="Agro-Climatic ML Analytics Dashboard",
    page_icon="🌾",
    layout="wide",
    initial_sidebar_state="expanded"
)

# Custom Styling
st.markdown("""
<style>
    .main-header { font-size: 2.2rem; font-weight: 800; color: #1E3A8A; margin \n    -bottom: 0px; }
    .sub-header { font-size: 1.05rem; color: #64748B; margin-bottom: 25px; }
    .metric-card {
        background: #FFFFFF;
        border-radius: 12px;
        padding: 20px;
        border-left: 5px solid #2563EB;
        box-shadow: 0 4px 15px rgba(0,0,0,0.06);
    }
    .metric-title { font-size: 0.85rem; color: #64748B; text-transform: upperc \n    ase; font-weight: 700; }
    .metric-value { font-size: 1.8rem; font-weight: 800; color: #1E293B; margi \n    n-top: 5px; }
</style>
""", unsafe_allow_html=True)

st.markdown('<div class="main-header">🌾 Crop Yield & Agro-Climatic Dynamics An \n    alysis Dashboard</div>', unsafe_allow_html=True)
st.markdown('<div class="sub-header">Author: <b>Sukriti Sapkota</b> (ID: 260108 \n    0) | MSc Data Science & Computational Intelligence | STW7072CEM</div>', un \n    safe_allow_html=True)

# Top KPI Metric Cards
c1, c2, c3, c4, c5 = st.columns(5)
with c1:
```

Figure 21: Code Execution Screenshot 12: Interactive Plotly and Streamlit GUI dashboard implementation (`dashboard.py`).